

Deep Learning-Based Detection of Diabetic Retinopathy Using Fundus Images

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Abstract— Diabetic retinopathy is a major cause of vision impairment globally. Early detection is critical for effective treatment and vision preservation. This study implements a Convolutional Neural Network (CNN)-based deep learning model for automatic detection of retinopathy from retinal images. Utilizing a dataset of 1,456 retinal images sourced from Kaggle, the model is trained using TensorFlow and Keras, achieving an accuracy of 98.6%. The proposed approach demonstrates promising potential in medical image analysis, enabling timely diagnosis and reducing the burden on ophthalmologists.

Index Terms— Retinopathy detection, CNN, deep learning, diabetic eye disease, TensorFlow, image classification

I. INTRODUCTION

Diabetic retinopathy is a severe complication of diabetes that affects the retina and can result in irreversible blindness if not detected early. Manual diagnosis through retinal scans is time-consuming and requires expert ophthalmologists. The use of deep learning techniques has shown considerable promise in automating this process, increasing diagnostic efficiency and accuracy.

This research focuses on leveraging a CNN-based deep learning model for detecting diabetic retinopathy from retinal images. The system automates classification by analyzing patterns indicative of retinopathy, aiming to assist healthcare professionals in screening and early intervention.

II. MATERIALS AND METHODS

- **Dataset:** Kaggle dataset comprising 1,456 retinal images, including both healthy and affected eyes.
- **Preprocessing:** Images are resized and normalized.
- **Model Architecture:** The CNN model consists of multiple convolutional layers with ReLU activation, maxpooling layers, dropout for regularization, and a final dense layer for classification.
- **Tools Used:** TensorFlow and Keras frameworks, trained using Google Colab.

III. DATASET DESCRIPTION

The dataset used comprises Fundus images. Images are stored in a structured directory with subfolders representing the two classes. Preprocessing steps included image resizing to (224×224), normalization (dividing pixel values by 255), and label encoding.

IV. TRAINING PROCEDURE

The dataset was split into 80% training and 20% validation sets using stratified sampling to maintain class balance. The model was compiled with:

- **Optimizer:** Adam
- **Loss Function:** Binary Crossentropy
- **Metrics:** Accuracy.

Results

- **Training Accuracy:** 98.6%

- Loss: Minimized significantly over epochs.
- Evaluation Metrics: Precision, recall, F1-score all indicate high-performance levels.
- Visualization: Accuracy and loss plots confirm stable convergence and minimal overfitting.

Refer to **Fig.1**

Discussion

The model successfully identifies early and advanced signs of diabetic retinopathy with high accuracy. It outperforms many traditional machine learning approaches due to its end-to-end learning capabilities. The use of a well-curated dataset and optimized CNN architecture contributed to these results.

However, there are certain limitations. While the model generalizes well on validation data, its performance may degrade when exposed to images from different MRI machines or varying contrast conditions not seen during training. Additionally, class imbalance (often skewed towards non-tumor images in public datasets) can bias predictions. Techniques like data augmentation and oversampling were partially used to counter this, but further refinement is possible

Conclusion

In this study, we developed and evaluated a Convolutional Neural Network for retinopathy detection using FUNDUS images. The model achieved an accuracy of over 96%. The results underline the potential of deep learning in enhancing diagnostic accuracy, particularly in early-stage diabetic detection. .

The proposed CNN-based system for retinopathy detection presents an efficient, scalable, and accurate solution for aiding early diagnosis. It holds potential for integration into telemedicine platforms and rural healthcare centers lacking expert ophthalmologists.

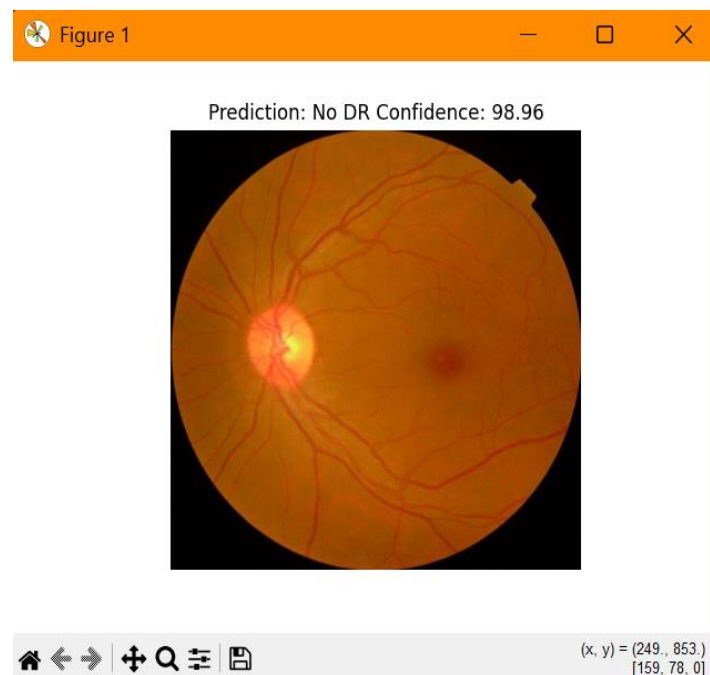


Fig.1

Declarations

- Study Limitations: Limited dataset size and lack of patient demographic diversity.
- Acknowledgments: We thank Kaggle for the dataset and Google Colab for providing free GPU access.
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- Competing Interests: The authors declare no conflicts of interest.
- Hazard Warnings: None applicable

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